Question 1:

Write MATLAB commands to define:

1. A row vector A containing the numbers 2, 4, 6, 8, 10.
2. A column vector B containing all odd numbers from 1 to 201.

Question 2:

1. Use the colon operator (:) to create a vector X starting at 0, incrementing by 0.5, and ending at 5.
2. Use the linspace function to create a vector Y containing exactly 100 points between 0 and π.

Question 3: Given the matrix $M = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$, write commands to:

1. Extract the element at row 2, column 3.
2. Extract the entire second row.
3. Extract a 2×2 sub-matrix from the top-right corner ([[20 30]; [50 60]]).

Question 4: Explain the difference between A * B and A .* B in MATLAB using two 2×2 matrices. What error might occur if their dimensions do not match?

Question 5: Write commands to generate:

1. A 3×3 identity matrix.
2. A 4×2 matrix filled entirely with zeros.
3. Find the dimensions (rows and columns) of a matrix $D = [123; 456]$ using a built-in function.

Question 6: Write a MATLAB script to plot the function $y = \sin(x)$ for x ranging from 0 to 2π with 200 points. Add an x-axis label ("Angle (rad)"), a y-axis label ("Sine Value"), and a title ("Plot of Sine Wave").

Question 7: Plot both $y_1 = \sin(x)$ and $y_2 = \cos(x)$ on the same set of axes for $x \in [0,2\pi]$. Use hold on to combine them, and include a legend to distinguish the two curves.

Question 8: Create a single figure window containing two subplots arranged vertically (2×1 grid):

1. Top subplot: Plot $y = x^2$ for x from -5 to 5.
2. Bottom subplot: Plot $y = e^x$ for x from 0 to 2.

Question 9: Generate data for $y = x^3$ where $x = -3{:}\,1{:}\,3$. Plot the graph using green circles (o) for markers, connected by a dashed green line, with a marker size of 8.

Question 10: Instead of a line plot, use MATLAB to visualize discrete data:

1. Create a bar chart for categorical sales data: sales = [120, 85, 150, 95] representing quarters Q1 to Q4.
2. Create a scatter plot for two random data vectors X and Y of length 50.